

An Integrality Result

Thm: $P = \{Ax = b\}$, $A, b \in \mathbb{Q}$ ~~$\mathbb{Z}$~~

has all integral vertices, iff
each rat'l supp. hyperplane
contains an integral vector.

" $\Rightarrow$ " obvious

" $\Leftarrow$ " last lecture.

An Integrality Result

Thm: $A, b \in \mathbb{Q}$, $\left[\begin{array}{l} \text{max: } c^T x \\ \text{st: } Ax \leq b \end{array} \right]$

has an integral opt value $\forall c \text{ integral}$

then $\{Ax \leq b\}$ is integral.

no: $\{Ax \leq b\}$ integral $\Rightarrow$ assumption - obvious

if: $H = \{c^T x = \delta\}$ a rational supporting hyperplane.

wlog assume that c is integral with rel. prime entries.

Enough to show, thanks to prev theorem, that H contains an integral vector.

$$H \text{ supp. hyp} \Leftrightarrow H \cap P = \operatorname{argmax}_{\text{st: } x \in P} c^T x$$

$\Rightarrow \delta$ is integral

$\Rightarrow H = \{x: c^T x = \delta\}$ must have an integral

soln, since $c \in \mathbb{Z}^n$ & rel prime, & δ integral. $\blacksquare$



Total Dual Integrality (TDI)

Hence: If $\max: c^T x$
 $\text{st: } Ax \leq b$ has integral
opt value & integral c , then ^{all} vertices are
integral.

In particular: If $\left[\begin{array}{l} \min: y^T b \\ \text{st: } y^T A = c^T \\ y \geq 0 \end{array} \right]$ has an integral
~~rational~~ solution

& integral c , for which the min is finite,

then primal polytope is integral.

Defn: (A, b) TDI if ^{dual} of $\left[\begin{array}{l} \max: c^T x \\ \text{st: } Ax \leq b \end{array} \right] = \left[\begin{array}{l} \min: y^T b \\ \text{st: } y^T A = c^T \\ y \geq 0 \end{array} \right]$ has
an integral solution & c integral for which RHS is finite.

$\forall b$ integral,

$A \text{ TU} \Rightarrow \underline{(A,b)} \Rightarrow \underline{\text{TDI.}}$

" $\Leftarrow$ " need not hold.

